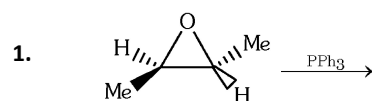


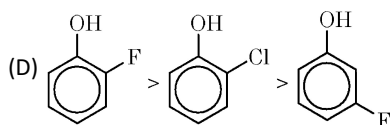
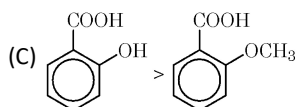
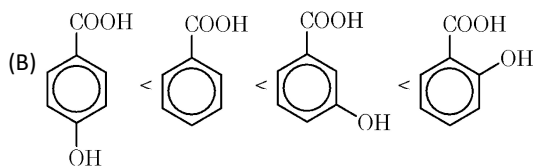
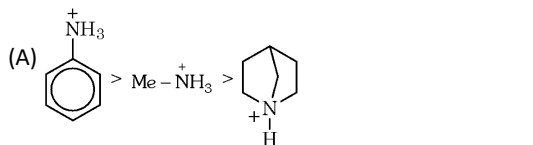
SCQ (Single Correct Type) :



Find product of above reaction is :

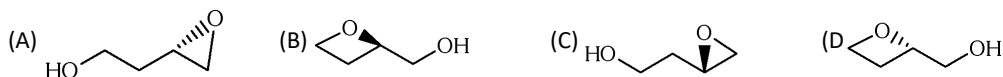
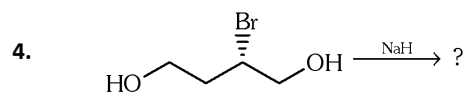


2. Which of the following order of acidic strength is incorrect.



3. On reductive ozonolysis of a polymer 4-oxopentanal is obtained. Find the monomer of polymer.

- (A) Penta-1,3-diene (B) 2-Methyl buta-1,3-diene
(C) 2-Methyl-2-butene (D) Penta-1,4-diene

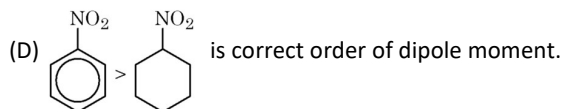
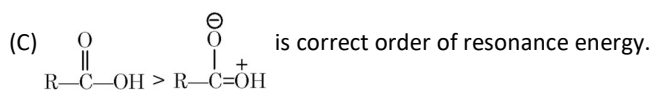
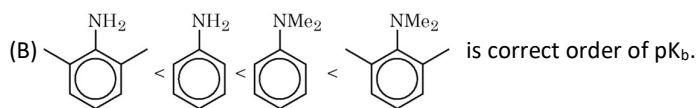
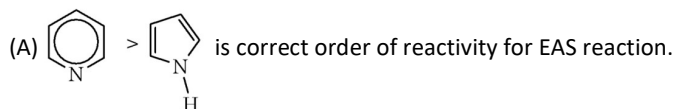


5. Select correct statements for following reaction :



- (A) 1st reaction gives only one organic product as result of syn addition.
 (B) For both reaction, 2 fractions of organic products are obtained.
 (C) 1st reaction gives mixture of two enantiomers as major product.
 (D) IInd reaction gives mixture of two enantiomers as major product.

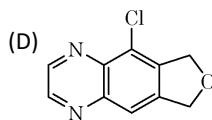
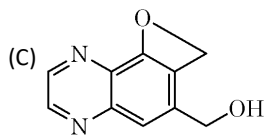
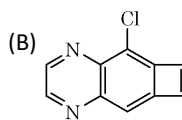
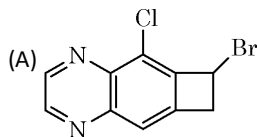
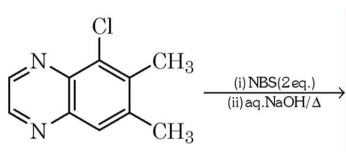
6. Select correct statement(s).



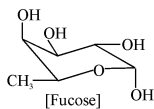
7. Which of the following statement is correct.

- (A) For butane-2, 3-diol, meso isomer is more stable than chiral isomers.
 (B) For 2, 3, 4, 5-tetramethyl hexane, chiral isomer is less stable than meso compound.
 (C) For chiral isomer 3, 4-dimethyl hexane, one conformations with zero dipole moment is possible.
 (D) Chiral isomer of butane-1, 2-diol may have a symmetrical conformation.

8.

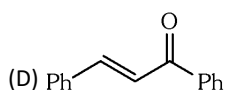
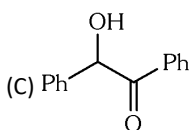
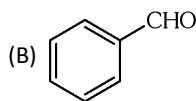
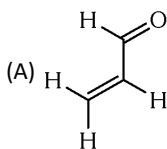


9. Select the incorrect statements about Fucose, a carbohydrate with the following structure.

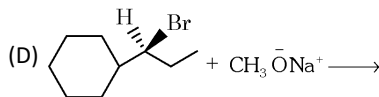
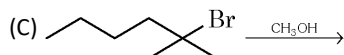
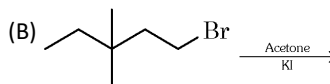
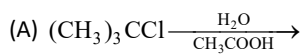


- (A) This is the L-enantiomer of fucose
(B) This is the α -anomer
(C) This is ketose
(D) It is a reducing sugar

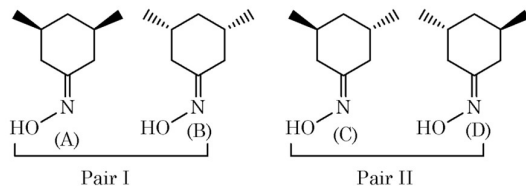
10. Positive Tollen's test is observed for



11. Among the following those which go faster when concentration of nucleophile is increased are:



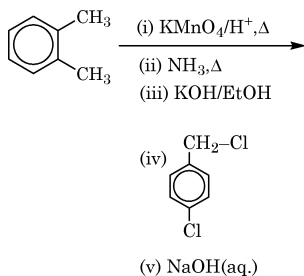
- 12.



Select correct statement for above pairs.

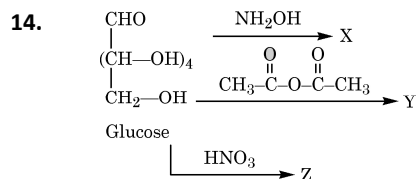
- (A) Compound A and B are geometrical isomers
(B) Compound C and D are enantiomeric compounds
(C) Compound A and B are not diastereomers
(D) Compound C and D are not geometrical isomers

- 13.



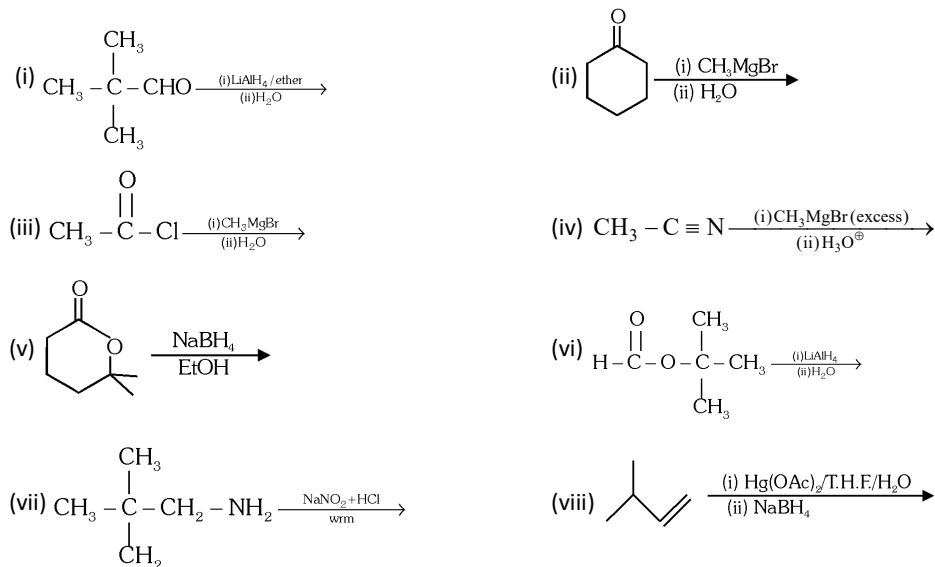
Identify the statement(s) about final nitrogen containing compound (product).

- (A) It is more basic than aniline.
- (B) Lone pair of nitrogen is involved in resonance.
- (C) It will give bad smelling compound on treatment with CHCl_3/KOH .
- (D) On treatment with $\text{NaNO}_2 + \text{HCl}$ followed by β -naphthol in alkaline medium it will form red dye.

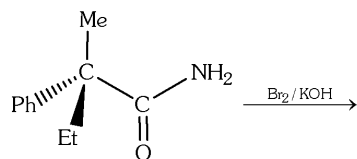


Identify true statement(s).

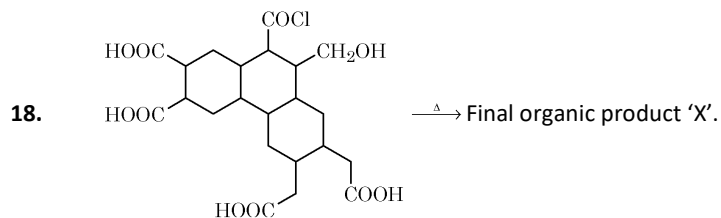
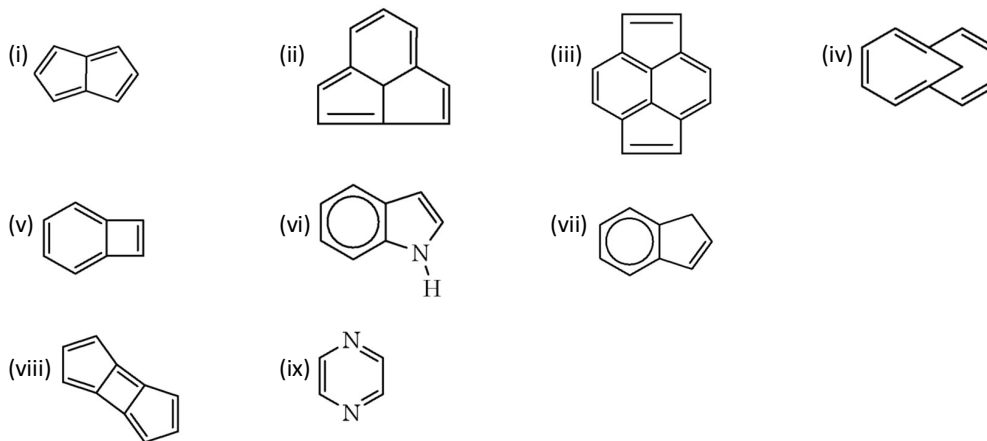
- (A) Y reacts with hydroxylamine to form oxime
 - (B) X contains the mixture of diastereomers
 - (C) Z is a dicarboxylic acid
 - (D) X contains two compounds of different boiling point
15. In how many of the following cases atleast one of final product is 3° alcohol



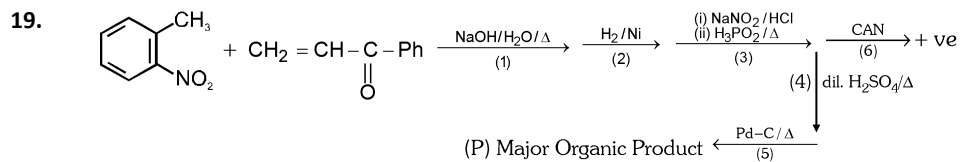
16. Number of organic products obtained in following reaction.



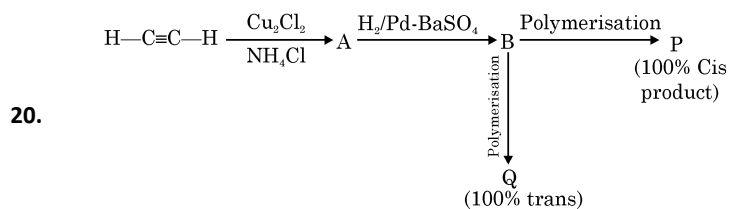
17. How many of the following compounds are not aromatic.



How many moles of LiAlH_4 are required to reduce 2 mole of X.



The product 'P' is having IHD ?



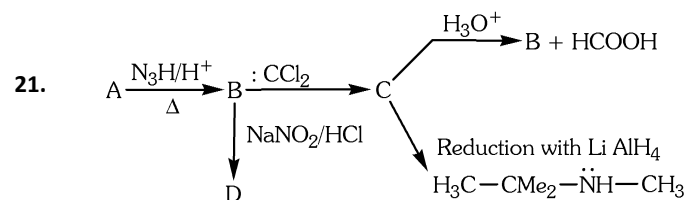
(i) 'P' is having more elongation property than "Q".

(ii) 'Q' is having more elongation property than 'P'.

(iii) $\text{B} \xrightarrow{\text{HCl}} \xrightarrow{\text{Polymerisation}}$ gives fibre

(iv) 'A' does not respond to $\text{NaNH}_2/\text{liq. NH}_3$

The number of incorrect statement(s) is/are.



Select number of correct statement/s among the following.

- (i) All compound A to D have same no. of Carbon atoms.
- (ii) B gives Hinsberg test
- (iii) D gives Hinsberg test
- (iv) D gives oxidized product with $\text{KMnO}_4/\text{H}^+/\Delta$
- (v) A gives red color with neutral FeCl_3
- (vi) B and C both give lassiagne test for nitrogen.

Answer Key

- | | | | | | | |
|---------|---------|-----------|-----------|------------|----------|-----------|
| 1. (B) | 2.(D) | 3. (B) | 4. (C) | 5. (D) | 6. (D) | 7. (B) |
| 8. (D) | 9. (C) | 10. (ABC) | 11. (BD) | 12. (ABCD) | 13. (AC) | 14. (BCD) |
| 15. (4) | 16. (1) | 17. (3) | 18. (3.5) | 19. (11) | 20. (3) | 21. (3) |